

Persistent homology on an observed galaxy field: discretization and survey-geometry systematics, a like-for-like framework, and a topological anomaly in DESI BGS

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Abstract. We apply three-dimensional persistent homology (PH) to the observed DESI Bright Galaxy Survey DR1 galaxy field (NGC; 217,614 galaxies, $z \in [0.1, 0.4]$) and show that naive PH comparisons between a bounded survey and periodic-box mocks are dominated by two systematics that can masquerade as physical anomalies. First, the mean H_1 persistence depends on the dimensionless smoothing parameter $\sigma_{\text{px}} = R/\Delta x$ rather than on the physical scale R alone ($+8.42\sigma$ between $\sigma_{\text{px}} = 0.216$ and 0.640 on matched fields), invalidating cross-survey comparisons at fixed R . Second, embedding a masked survey in a zero-padded cube and computing an unmasked filtration generates boundary-driven features: a spuriously high mean persistence together with a suppressed generator count — a “dual signature” that is largely persistence-diagram truncation, not cosmology. We introduce a like-for-like framework — cut-sky mocks with matched footprint, radial selection, redshift-space distortions, galaxy density, weighting, and σ_{px} , analysed with a masked filtration — and apply it to DESI BGS against 2000 Quijote w_0 CDM cosmologies ($w_0 \in [-1.30, -0.70]$, $w_a = 0$). Most of the apparent signal dissolves, including the mean-persistence anomaly and its apparent w_0 sensitivity. A single feature survives: the DESI field contains fewer H_1 generators than all 2000 like-for-like mocks (empirical $p < 1/2000$), robust to an HOD refit to the observed projected clustering, to weighting, and to a density-dependent incompleteness surrogate; no $w_0 \in [-1.30, -0.70]$ reproduces it (extrapolation requires $w_0 \approx -294$), formally excluding w_0 CDM with $w_a = 0$ as its origin. We report this deficit as an open anomaly — explicitly not a dark-energy detection — and release the pipeline as a template for PH analyses of observed galaxy fields.

Keywords: cosmological simulations, redshift surveys, cosmological parameters from LSS, galaxy clustering

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1 Introduction

Persistent homology (PH) provides a mathematically rigorous, multiscale characterization of the topology of the cosmic web — connected components (β_0), filamentary loops (β_1), and voids (β_2) — and has emerged as a promising beyond-two-point statistic for cosmological inference. Fisher forecasts on dark-matter halo catalogues show that PH summaries can outperform the power spectrum for Ω_m and σ_8 [1]; applications to Dark Energy Survey Year 3 convergence maps deliver a $\sim 70\%$ figure-of-merit improvement over two-point statistics for $S_8\text{--}\Omega_m$ [2]; and machine-learning pipelines built on persistence images enable field-level parameter inference on simulations [3]. The sensitivity of PH statistics to redshift-space distortions has likewise been quantified on simulated fields [4].

What this literature has in common is that it operates on simulated periodic boxes or on two-dimensional projected maps. To our knowledge, three-dimensional PH had not previously been applied to an *observed* spectroscopic galaxy field — a bounded, masked, flux-limited region with a radial selection function, observational weights, and redshift-space positions, voxelized at a resolution set by the survey volume rather than by choice. This paper does so, using the DESI Bright Galaxy Survey (BGS) DR1 [5, 6] as the test field, and reports that the exercise is dominated by systematics that the simulation-based and projected-map literature has not had to confront. Left uncontrolled, these systematics produce apparent topological anomalies of very high formal significance that have nothing to do with cosmology.

1.1 This work: a research trajectory and its methodological yield

This work originated as a targeted search: whether the topology of the galaxy density field could constrain dynamical dark energy, motivated by the DESI DR2 evidence for an evolving equation of state [5, 7]. We note for accuracy that the DESI DR2 preferred solution in the Chevallier–Polarski–Linder (CPL) parametrization lies in the quadrant $w_0 > -1$, $w_a < 0$ (best-fit $w_0 = -0.838 \pm 0.086$ for DR2+CMB), i.e. a dark energy that was phantom-like in the past and quintessence-like today; the present analysis, whose mock suite is restricted to $w_a = 0$, cannot test that solution (Section 2.3).

An initial exploratory comparison between the DESI BGS field and halo-occupation-populated Quijote mocks [8] yielded a topological anomaly of apparently very high significance, accompanied by an anti-correlated “dual signature” (high mean H_1 persistence, suppressed generator count). Pursuing that signal to the standard it would need for a cosmological claim required building a progressively stricter comparison — matching the survey footprint, radial selection, redshift-space treatment, galaxy density, weighting, and grid discretization between data and mocks, and treating the survey boundary consistently in the filtration. Each tightening of the comparison isolated a systematic of general interest, and most of the original signal was reabsorbed by them. We consider this trajectory — an initial detection claim dismantled by the very tools built to defend it — to be the central result, and we document it explicitly (Appendix D) rather than presenting only the final pipeline.

The paper makes four contributions:

1. **The σ_{px} systematic** (Section 3.2). The mean H_1 persistence of a voxelized field depends on the dimensionless smoothing parameter $\sigma_{\text{px}} = R/\Delta x$, not on the physical smoothing scale R alone. On matched mock fields the shift between $\sigma_{\text{px}} = 0.216$ and 0.640 is $+8.42\sigma$ of the mock scatter — larger than any cosmological signal in the suite. Cross-survey PH comparisons at fixed R but different voxel size are therefore systematically biased, and σ_{px} matching should become a standard diagnostic.

2. **Survey-geometry and boundary artefacts** (Section 3.3). Embedding a masked survey (here filling 14.7% of the bounding cube) in a zero-padded volume and computing an unmasked cubical filtration manufactures a dual signature: boundary-spanning generators inflate the mean persistence while the reduced effective volume suppresses the generator count. The anti-correlated pattern that superficially resembles a physical signal is largely the fingerprint of a truncated persistence diagram.
3. **A like-for-like framework** (Section 4). We construct cut-sky mocks — periodic boxes tiled into the survey embedding cube, carved by the DESI angular mask and $n(z)$, placed in redshift space, density- and weighting-matched, voxelized against the survey random catalogue, and analysed with a masked filtration that excludes the exterior on both sides. This is, we argue, the minimal construction under which a PH comparison between an observed field and simulations is meaningful.
4. **A case study with an honest residual** (Section 5). Applied to DESI BGS, the framework dissolves the mean-persistence anomaly and the apparent w_0 trend. One feature survives: the observed field contains $\sim 20\%$ fewer H_1 generators than every one of the 2000 like-for-like Λ CDM/ w_0 CDM mocks (empirical $p < 1/2000$). The deficit is unchanged by an HOD refit to the observed projected clustering, by the observational weighting, and by a density-dependent incompleteness surrogate; and no w_0 in the sampled range approaches it, excluding w_0 CDM ($w_a = 0$) as its origin. We report it as an open anomaly and enumerate what would be required to explain or eliminate it.

Throughout, we deliberately quote empirical ranks rather than Gaussian-tail significances wherever the mock count sets a resolution floor, and we flag every post-hoc analysis as such. The pipeline, frozen numerical records, and all scripts are publicly available (Data Availability).

2 Data

2.1 DESI BGS DR1

We use the DESI Bright Galaxy Survey Data Release 1 (BGS DR1) large-scale structure catalogues, specifically the North Galactic Cap (NGC) footprint, comprising 217,614 galaxies at $z \in [0.1, 0.4]$ with effective redshift $z_{\text{eff}} \approx 0.2$. Galaxy weights $w = w_{\text{sys}} \times w_{\text{comp}} \times w_{\text{FKP}}$ are applied during mass assignment. The accompanying random catalogue (13,248,857 randoms) defines the survey geometry and radial selection on the analysis grid, and is used both for the FKP-style density estimate and for the projected clustering measurement $w_p(r_p)$ (TreeCorr [9], Landy–Szalay). The South Galactic Cap is retained as a cross-check.

Galaxy and random positions are converted to comoving coordinates assuming the Quijote fiducial cosmology ($\Omega_m = 0.3175$, $h = 0.6711$ [8]), which is Planck-motivated [10] and matches the simulation side of the comparison; sensitivity of the voxelized field to the assumed fiducial (Alcock–Paczynski-type distortion) is not assessed here and is listed as a limitation. Positions are voxelized onto a 128^3 grid spanning the survey bounding cube ($L = 1997.4 h^{-1}\text{Mpc}$, cell size $\Delta x_{\text{DESI}} = 15.60 h^{-1}\text{Mpc}$) with cloud-in-cell (CIC) weighting. The survey occupies 307,805 of the 128^3 voxels (14.7% of the embedding cube), a fact that drives much of Section 3.3.

2.2 Quijote nwLH mock cosmologies

We use the 2000 cosmologies of the Quijote nwLH Latin Hypercube [8], which samples the seven parameters Ω_m , Ω_b , h , n_s , σ_8 , M_ν , and w_0 , with $w_0 \in [-1.30, -0.70]$ at fixed $w_a = 0$, each providing Friends-of-Friends halo catalogues in a periodic box of side $1000 h^{-1} \text{Mpc}$. Throughout this work we use the $z = 0.5$ snapshot (`groups_003`), the closest available to the BGS effective redshift; the residual snapshot-to-lightcone growth mismatch ($z_{\text{eff}} \approx 0.2$ vs. $z = 0.5$) is declared as a limitation (Section 7). Galaxy catalogues are generated with the five-parameter halo occupation distribution (HOD) of Zheng et al. [11]; central galaxies carry the host-halo bulk velocity and satellites additionally receive an isotropic virial velocity dispersion in the final (masked) pipeline. Halo catalogues are restricted to $N_{\text{part}} \geq 20$ ($M_h \gtrsim 8.6 \times 10^{12} h^{-1} M_\odot$ at this resolution), which lies safely below all occupation thresholds employed.

2.3 Structural limitation: fixed $w_a = 0$

The nwLH suite varies w_0 at fixed $w_a = 0$. The CPL solution preferred by DESI DR2 combinations ($w_0 = -0.667$, $w_a = -1.079$ for DR2+CMB+DES-Y5 [5]) is therefore not testable with the present simulations. Every statement in this paper about dark energy is consequently restricted to the $w_0\text{CDM}$ family with $w_a = 0$: we will show that this family is excluded as the origin of the surviving anomaly (Section 5.4), but the exclusion does not extend to the full CPL space.

3 Methods I: the TDA pipeline and its systematics

3.1 Pipeline definition

The density contrast is estimated with the FKP-style ratio against the random field,

$$\delta(\mathbf{x}) = \frac{n_{\text{data}}(\mathbf{x}) - \alpha n_{\text{rand}}(\mathbf{x})}{\alpha n_{\text{rand}}(\mathbf{x})}, \quad \alpha = \frac{\sum w_{\text{data}}}{\sum w_{\text{rand}}}, \quad (3.1)$$

on the 128^3 grid, with $\delta = 0$ assigned outside the survey mask. The field is then log-transformed *on the raw density contrast*,

$$\nu(\mathbf{x}) = \log(1 + \text{clip}(\delta, -1)), \quad (3.2)$$

smoothed with a Gaussian kernel of width R , and mean-subtracted within the survey mask. The ordering matters: the log-transform must precede smoothing and mean subtraction, both because $\delta \geq -1$ holds only for the raw contrast and because the transform is what bounds the FKP shot-noise tail (Section 4.1). We emphasise this because an earlier version of this pipeline *stated* Eq. (3.2) but did not apply it; restoring it shifts the mean H_1 persistence of the DESI field by a factor ~ 2 at fixed grid (Appendix D), which by itself disqualifies $\langle \text{pers}_1 \rangle$ as a robust cross-pipeline statistic unless the transform is specified and verified.

Persistent homology of the superlevel filtration of ν is computed with the `CubicalComplex` implementation of the `gudhi` library [12]; births and deaths follow the superlevel convention $b = -\text{diag}[:, 0]$, $d = -\text{diag}[:, 1]$, persistence $p = b - d > 0$. Eight scalar features are extracted per field (Appendix A); the two relevant here are the finite H_1 generator count at the Betti-curve peak, β_1^{max} , and the mean finite H_1 persistence,

$$\langle \text{pers}_1 \rangle = \frac{1}{|D_1|} \sum_{(b,d) \in D_1} (b - d). \quad (3.3)$$

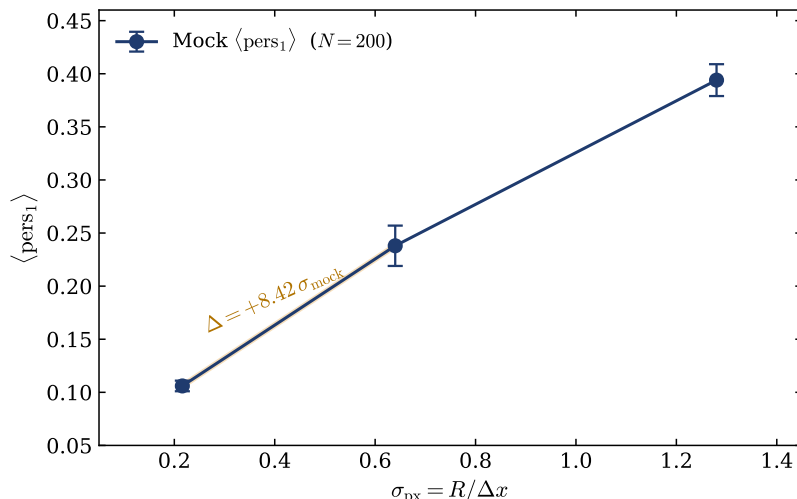


Figure 1. Dependence of the mock mean H_1 persistence on the dimensionless smoothing parameter $\sigma_{\text{px}} = R/\Delta x$. The shift between the two σ_{px} values relevant to a naive DESI–Quijote comparison exceeds the full w_0 lever arm of the suite by more than an order of magnitude.

β_1^{max} is a connectivity (counting) statistic; $\langle \text{pers}_1 \rangle$ is an amplitude statistic. This distinction, introduced here informally, is quantified in Section 5.3 and turns out to govern which features survive systematic control.

3.2 The σ_{px} systematic

PH on a voxelized field is sensitive to the dimensionless smoothing parameter $\sigma_{\text{px}} = R/\Delta x$ — the kernel width in *voxel units* — rather than to the physical scale R alone, because σ_{px} controls the inter-voxel correlation length of the discretized field: at fixed physical R , a coarser grid produces more transient low-persistence generators, a finer grid fewer but more persistent ones. We quantify this on $N = 200$ nwLH mock fields at three values of σ_{px} (Table 4, Appendix B): the difference between $\sigma_{\text{px}} = 0.216$ and 0.640 is $\Delta \langle \text{pers}_1 \rangle = +0.153$, i.e. $+8.42\sigma$ of the mock scatter — larger than the entire w_0 lever arm in the suite by more than an order of magnitude.

The consequence for observational work is immediate. A survey field and a simulation box generally have different voxel sizes (15.60 vs. $7.81 h^{-1}\text{Mpc}$ here), so a comparison at fixed physical R compares different σ_{px} and inherits an $\mathcal{O}(8\sigma)$ bias; conversely, matching σ_{px} by smoothing the two fields at different physical scales erases real structure on one side only, trading a discretization systematic for a physical-scale one. The only clean configuration is the one adopted in Section 4: place both fields on the *same* grid with the *same* physical R , so that σ_{px} is matched by construction rather than by compensation. We recommend that any PH analysis comparing fields voxelized at different resolutions report σ_{px} explicitly and demonstrate matching.

3.3 Survey-geometry and boundary artefacts: the anatomy of a false dual signature

The DESI BGS NGC region fills 14.7% of its bounding cube. A superficially natural procedure — set the exterior to $\delta = 0$, compute the cubical filtration on the full cube, and

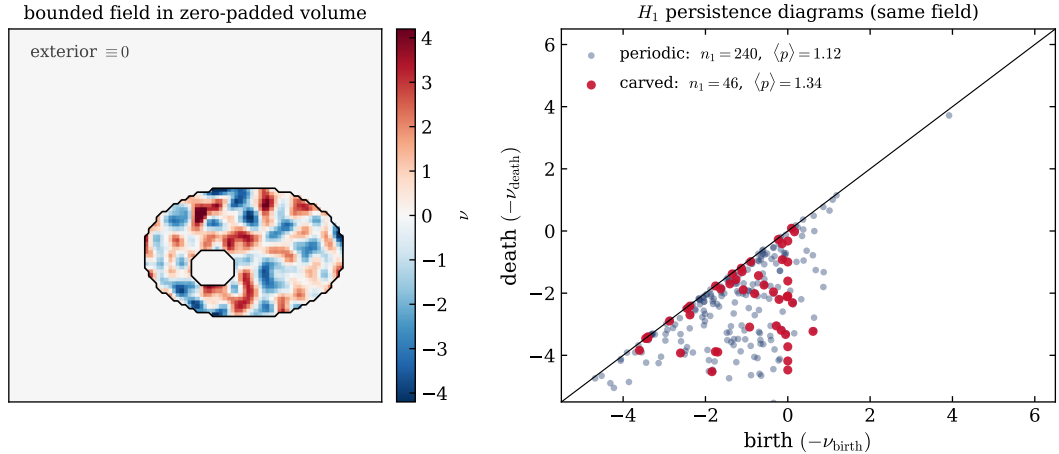


Figure 2. Anatomy of the boundary artefact, demonstrated on a two-dimensional Gaussian random field (an illustration, not survey data; the persistence computation is real). Left: the field carved by a survey-like mask with a veto hole, embedded in a zero exterior and mean-subtracted in the mask — the configuration of a naive observational analysis. Right: finite H_1 persistence diagrams of the identical field analysed as a periodic domain (blue) versus carved (red): the generator count collapses ($240 \rightarrow 46$) while the mean persistence *rises* ($1.12 \rightarrow 1.34$). The anti-correlated high-persistence/low-count pattern — the “dual signature” — is manufactured by diagram truncation alone.

compare against periodic-box mocks — was the configuration of our own initial analysis, and it produces two large, anti-correlated apparent anomalies:

- *Suppressed generator count.* Loops require volume to form and are cut wherever they would cross the survey boundary. A field occupying a small, irregular fraction of the complex trivially contains far fewer H_1 generators than a periodic box. In our initial configuration the observed suppression was a factor ~ 4.5 ($\beta_1^{\max} = 29,683$ vs. a periodic-mock mean of $133,786$), formally $\sim -56\sigma$. That number measures the mask, not the sky: the NGC and SGC generator counts scale with their respective in-survey voxel counts to within 4% (voxel ratio 1.79, generator ratio 1.86), which is the signature of a volume-driven, not cosmology-driven, effect.
- *Inflated mean persistence.* The interface between the survey fluctuations and the flat zero exterior seeds generators that span a large fraction of the filtration range; simultaneously, removing many low-persistence generators from the diagram (the previous item) raises the *mean* persistence of the survivors by construction. High mean persistence with a low count — a pattern tempting to read as “fewer but more robust cosmic-web loops” — is precisely the fingerprint of a truncated persistence diagram.

The joint lesson is that an anti-correlated dual anomaly in $(\langle \text{pers}_1 \rangle, \beta_1^{\max})$, however physically suggestive, is the *expected* outcome of comparing a bounded, zero-embedded field against periodic boxes, and carries no cosmological information until the geometry is matched. The quantitative dissolution of our own dual signature under geometry matching is presented in Section 5.1.

4 Methods II: the like-for-like framework

The framework rests on a single principle: every operation applied to the data field is applied, identically and by the same code path, to the mock fields. Concretely, for each of the 2000 nwLH cosmologies:

1. *Galaxy catalogue.* Haloes ($z = 0.5$, $N_{\text{part}} \geq 20$) are populated with the Zheng et al. HOD; centrals inherit the halo bulk velocity, satellites are placed uniformly within the virial radius and receive the halo bulk velocity plus an isotropic virial dispersion $\sigma_{1D}(M) = \sqrt{GM/3R_{\text{vir}}}$.
2. *Cut-sky construction.* The periodic box is tiled into the DESI embedding cube; an observer is placed at the coordinate origin; angular positions and cosmological redshifts follow, with $z_{\text{obs}} = z_{\text{cosmo}} + (1 + z_{\text{cosmo}}) v_{\parallel}/c$ placing the catalogue in redshift space. Galaxies are kept if their embedding voxel lies inside the DESI NGC mask and $z_{\text{obs}} \in [0.1, 0.4]$.
3. *Selection matching.* A global radial downsampling reproduces the BGS $n(z)$ shape and fixes the selected count at the BGS NGC total ($N_{\text{sel}} \simeq 217,614$; achieved median 217,570).
4. *Field construction.* The mock counts are voxelized by the same CIC routine and divided against the *same* DESI random field (Eq. 3.1); Eq. (3.2) is applied to the raw contrast; the field is smoothed at $R = 5 h^{-1}\text{Mpc}$ on the DESI grid ($\sigma_{\text{px}} = 0.320$, matched to the data by construction) and mean-subtracted in the mask.
5. *Masked filtration.* The exterior is excluded from the persistence computation on both sides: exterior cells are assigned a sentinel value that enters the filtration last, and any generator whose birth or death touches the sentinel is discarded. Loops crossing the survey boundary are thereby cut identically for data and mocks — the topologically consistent treatment of a bounded survey.

Steps 1–5 remove, by construction, the σ_{px} mismatch (Section 3.2), the boundary artefact (Section 3.3), the real-space/redshift-space mismatch, and the density mismatch. Two residual asymmetries — the observational weights carried by the data and the fidelity of the HOD — are addressed empirically in Section 5.3.

4.1 Field amplitude and the role of the log-transform

The FKP estimator (Eq. 3.1) is numerically fragile where the random coverage is low: mock galaxies falling in voxels of near-zero expected density generate contrast spikes up to $\delta \sim +150$. Without Eq. (3.2) the in-survey standard deviation of the mock fields exceeds that of the data by a factor ~ 31 , and any amplitude-sensitive statistic — $\langle \text{pers}_1 \rangle$ first among them — compares nothing but this mismatch. With the transform applied to the raw contrast on both sides, the amplitude ratio falls to 0.79 and the filtration ranges become commensurate. We regard the verified presence and placement of the log-transform as part of the like-for-like definition, not an implementation detail.

5 Results: the DESI BGS case study

We now apply the framework to DESI BGS NGC. The data-side reference values, recomputed through the identical masked pipeline, are $\langle \text{pers}_1 \rangle = 0.7246$ and $\beta_1^{\text{max}} = 28,256$ for the DESI field. Throughout this section, significances are quoted primarily as empirical ranks within the mock distribution; with N mocks the smallest measurable p -value is $1/(N+1)$, and Gaussian-equivalent z -scores beyond that are tail extrapolations which we report only parenthetically.

5.1 What dissolves

The dual signature. Under the naive configuration (Section 3.3) the generator-count suppression was a factor ~ 4.5 ($\sim -56\sigma$ formally). Under the like-for-like construction the mock count falls from 133,786 (periodic) to $35,437 \pm 313$ (cut-sky, masked): survey geometry alone absorbs the large majority of the apparent suppression. The anti-correlated pattern as a *signature* — the joint high-persistence/low-count anomaly — does not survive (see next paragraph); what remains is a single-feature deficit discussed in Section 5.2.

The mean-persistence anomaly. The DESI value $\langle \text{pers}_1 \rangle = 0.7246$ against the masked cut-sky mock distribution 0.6299 ± 0.0439 gives an empirical rank of 98.0% ($z = +2.16$): inside the mock distribution, not significant. The progression is instructive: the apparent $\langle \text{pers}_1 \rangle$ excess was $+6.65\sigma$ -equivalent under the exact-replica (unmasked) cut-sky test, weakened to a worst-case $+4.67$ under weighting variations, and fell below 3σ once boundary-crossing generators were excluded from the filtration. A statistic that loses significance monotonically as the comparison is made more correct is a systematic residue, not a signal, and we retire $\langle \text{pers}_1 \rangle$ accordingly. Its $\sim 2\times$ sensitivity to the log-transform (Section 3.1) independently supports this classification.

The apparent w_0 sensitivity. In the naive configuration both features appeared to correlate with w_0 in directions consistent with a phantom interpretation. Under the like-for-like construction the partial correlation of β_1^{max} with w_0 (conditioned on Ω_m, σ_8) is $r = +0.015$ (permutation $z = +0.24$): compatible with zero. The apparent trend was an artefact of the mismatched comparison, and with it the phantom reading of our earlier analysis loses its basis. We consider this dissolution — obtained with our own tools, on our own prior claim — the strongest internal validation of the framework.

5.2 What survives: an H_1 generator deficit

One feature survives every control. The DESI field contains $\beta_1^{\text{max}} = 28,256$ finite H_1 generators at the Betti-curve peak, against a like-for-like mock distribution of $35,437 \pm 313$ ($N = 2000$): a $\sim 20\%$ deficit, with the observed value below *all* 2000 mocks — empirical $p < 1/2000$ ($\approx 3.3\sigma$ -equivalent at the resolution floor; the Gaussian extrapolation would be -22.9σ and we explicitly do not use it as the headline number, since the mock scatter does not include cosmic variance at the BGS volume, Section 5.5).

The deficit is not a smoothness effect in the naive sense. On the same ν fields, the data show *less* small-scale spectral power than the mocks (fractional power above half-Nyquist: 0.570 vs. 0.680 ± 0.007) but *more* high-density excursion (fraction of in-survey voxels above $\nu = 1$: 0.491 vs. 0.241 ± 0.036) and a *higher* gradient variance (7.80 vs. 4.98 ± 0.89). The one-point structure of the observed field is more concentrated than the mocks — heavier high-density tail, fewer diffuse small-scale fluctuations — and it is the diffuse fluctuations that seed H_1 generators. Whether this concentration reflects residual observational modelling

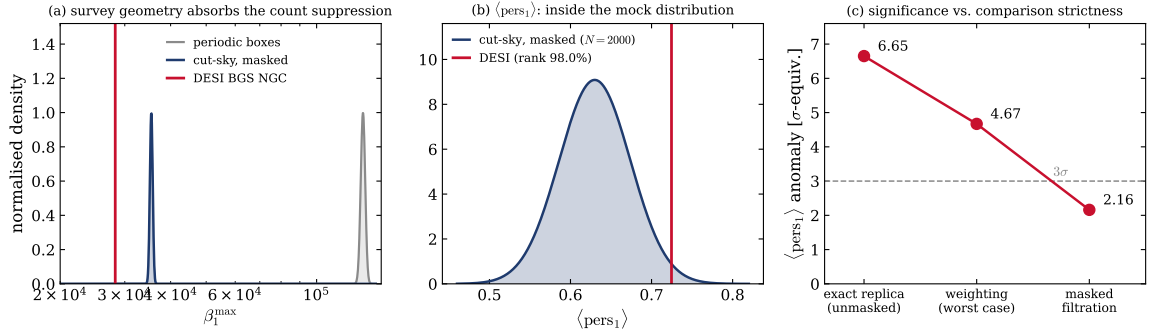


Figure 3. What dissolves under the like-for-like framework. (a) The generator-count comparison: moving from periodic boxes (grey) to cut-sky masked mocks (blue) absorbs the bulk of the apparent suppression relative to the DESI value (red). (b) The mean persistence: under the masked framework the DESI value lies inside the mock distribution (empirical rank 98.0%). (c) The $\langle \text{pers}_1 \rangle$ anomaly decreases monotonically as the comparison is made stricter — the behaviour of a systematic residue, not of a signal. Mock distributions in (a) and (b) are drawn as Gaussian densities from the frozen summary statistics (mean, scatter, N).

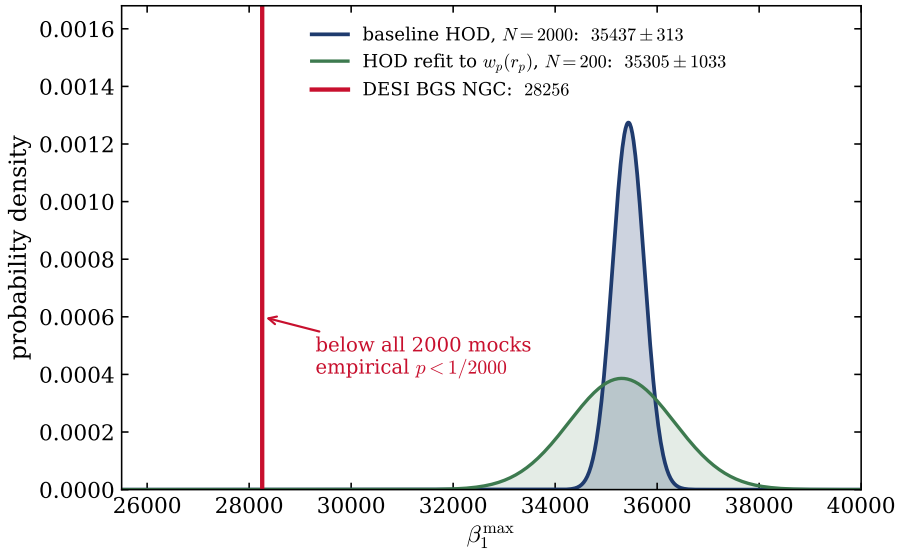


Figure 4. The surviving anomaly: the DESI BGS NGC H_1 generator count (red) lies below all 2000 like-for-like masked mocks (blue; empirical $p < 1/2000$) and below 199 of the 200 mocks built with the HOD refit to the observed projected clustering (green). Mock distributions are drawn as Gaussian densities from the frozen summary statistics (mean, scatter, N).

or a genuine field-level property is the open question; the exclusion battery below removes the candidates we could test.

5.3 The exclusion battery

Observational weighting. The data field carries $w_{\text{sys}}w_{\text{comp}}w_{\text{FKP}}$; the mocks are unweighted. We bound the effect from both directions: rebuilding the DESI reference with

Table 1. The exclusion battery for the $\beta_1^{\max}$ deficit. All comparisons use the like-for-like masked framework. Ranks are empirical; “ z ” denotes the Gaussian-equivalent tail extrapolation, reported for orientation only.

Control	Mock $\beta_1^{\max}$	DESI rank	Outcome
Baseline (B3 HOD), $N=2000$	35437 ± 313	0/2000	deficit present
Unit weights on data	$28256 \rightarrow 28565$	—	unchanged ($\sim 1\%$)
FKP-like weights on mocks	$35838 \rightarrow 36460$	—	moves away from data
HOD refit to $w_p(r_p) + \bar{n}$, $N=200$	35305 ± 1033	1/200	deficit present
Incompleteness surrogate, $f \leq 0.30$	$35467 \rightarrow 35582$	—	opposite sign
$w_0 \in [-1.30, -0.70]$ (Sec. 5.4)	no dependence	—	excluded

unit weights shifts $\beta_1^{\max}$ by $+309$ ($\sim 1\%$), and assigning the mocks an FKP-like positional weight shifts their mean by $+622$ ($\sim 2\%$) — in the direction *away* from the data. The worst-case separation across weighting choices remains -24.6σ -equivalent. $\beta_1^{\max}$, a connectivity statistic, is insensitive to weighting; $\langle \text{pers}_1 \rangle$, an amplitude statistic, is not (its worst-case separation drops to $+4.67$), which is the quantitative basis for the connectivity/amplitude distinction drawn in Section 3.1.

HOD calibrated to the observed clustering. The baseline B3 occupation ($\log M_{\min} = 12.5$, $\log M_1 = 13.5$, $\alpha = 1.0$) was not fit to the data. We measure $w_p(r_p)$ for BGS NGC with TreeCorr (Landy–Szalay estimator; $r_p \in [0.5, 30] h^{-1}\text{Mpc}$, $\pi_{\max} = 40 h^{-1}\text{Mpc}$) and fit $(\log M_{\min}, \log M_1, \alpha)$ jointly to $w_p(r_p)$ and the number density on fiducial-cosmology boxes. The best fit ($\log M_{\min} = 13.1$, $\log M_1 = 13.8$, $\alpha = 0.8$; Table 2) is *more* concentrated than the baseline — the direction that would, if anything, help reproduce a loop deficit. It does not: cut-sky mocks built with the fitted HOD give $\beta_1^{\max} = 35,305 \pm 1033$ ($N = 200$), statistically indistinguishable from the baseline mock mean, with the DESI value at empirical rank 0.5% (below 199 of 200). Even the occupation preferred by the data’s own clustering does not produce the deficit. We note the fit is a coarse-grid robustness calibration at $z = 0.5$, not a full posterior (Section 7).

Density-dependent incompleteness (fiber-assignment surrogate). DESI fiber assignment preferentially loses galaxies in angularly crowded regions. Lacking the DESI alternate-MTL mocks, we apply a surrogate: mock galaxies are dropped with probability proportional to their local smoothed density, normalised to mean drop fractions $f \in \{0.05, 0.15, 0.30\}$ — the last far exceeding realistic BGS incompleteness. The mock $\beta_1^{\max}$ does not move toward the data at any f ; it increases slightly (from 35,467 at $f = 0$ to 35,582 at $f = 0.30$), because density-dependent decimation fragments dense regions and *creates* small transient loops. Within this model, fiber-assignment incompleteness not only fails to explain the deficit but acts with the opposite sign. The test with the official alternate-MTL mocks remains future work, and we already note that the data-side completeness weights, whose removal leaves $\beta_1^{\max}$ unchanged (above), point the same way.

5.4 Exclusion of $w_0\text{CDM}$ ($w_a = 0$)

The suite spans $w_0 \in [-1.30, -0.70]$; if the deficit were a dark-energy effect within this family, the most phantom mocks should approach the data. They do not. The partial correlation $r(\beta_1^{\max}, w_0 | \Omega_m, \sigma_8) = +0.015$ is compatible with zero; the phantom ($w_0 < -1.10$) and quintessence ($w_0 > -0.90$) subsets have indistinguishable means ($35,411 \pm 302$ vs. $35,414 \pm$

Table 2. HOD fit to the observed projected clustering (coarse grid; χ^2 combines fractional w_p residuals and the number-density term). The best fit is more concentrated than the baseline, yet leaves the mock $\beta_1^{\max}$ unchanged.

	$\log M_{\min}$	$\log M_1$	α
Baseline (B3)	12.5	13.5	1.0
Best fit to $w_p(r_p) + \bar{n}$	13.1	13.8	0.8

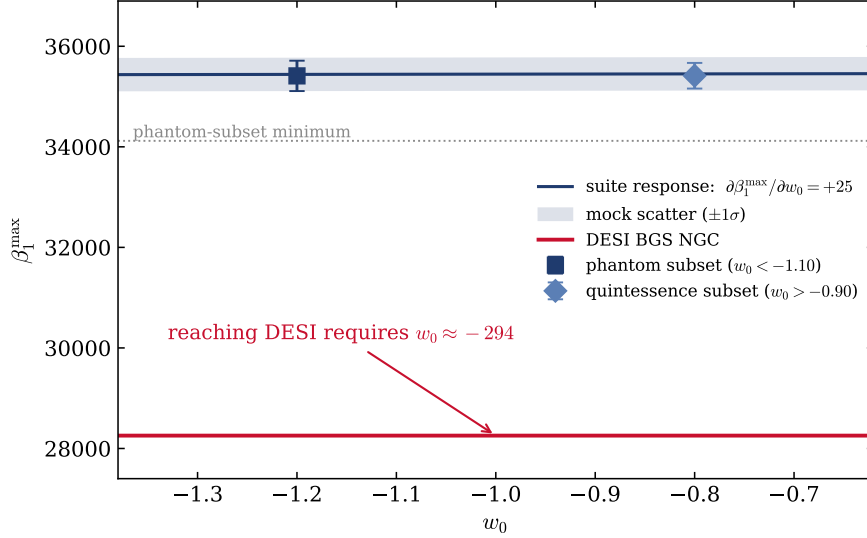


Figure 5. Exclusion of w_0 CDM ($w_a = 0$). The suite response of $\beta_1^{\max}$ to w_0 (blue line, $\partial\beta_1^{\max}/\partial w_0 = +25$ per unit w_0 , with the $\pm 1\sigma$ mock scatter band) is negligible: the phantom ($w_0 < -1.10$) and quintessence ($w_0 > -0.90$) subset means are indistinguishable, and the observed value (red) lies below the minimum of the phantom subset (dotted). Reaching the observation by extrapolating the response would require $w_0 \approx -294$.

255); and the observed value lies below the *minimum* of the phantom subset. A linear response model across the suite gives $\partial\beta_1^{\max}/\partial w_0 \approx +25$ per unit w_0 : reaching the observed value by extrapolation would require $w_0 \approx -294$. We conclude that no w_0 CDM model with $w_a = 0$ in — or anywhere near — the sampled range reproduces the deficit: **the anomaly is not a w_0 effect.**

This exclusion is, we believe, the correct role for the w_0 insensitivity of topological counting statistics that our earlier analysis had treated as a tension to explain away. It also sharpens the falsifiable content of the anomaly: whatever produces the deficit, it is not the expansion-history parameter our simulation suite varies. The CPL space with $w_a \neq 0$ remains untested (Section 2.3).

5.5 Significance, stated honestly

The defensible statement is: *the observed $\beta_1^{\max}$ lies below all 2000 like-for-like mocks, empirical $p < 1/2000$.* We do not promote the Gaussian-equivalent -22.9σ to a headline for two reasons. First, with $N = 2000$ mocks the empirical resolution floor is $p \simeq 1/2001$ ($\approx 3.3\sigma$); anything beyond is an extrapolation of tails we have not characterised. Second, the denomi-

nator is the mock-to-mock scatter of cut-sky realisations carved from $1000 h^{-1}\text{Mpc}$ boxes; it does not include cosmic variance at the BGS volume on the data side, nor covariance with the data-side measurement, and a complete error budget would require volume-matched independent lightcones. Both caveats act in the direction of reducing formal significance, and we adopt the rank statement accordingly. For the same reason we quote the earlier, superseded significances of this project’s initial analysis ($+5.96\sigma$ to $+29.9\sigma$, the last obtained by post-hoc conditioning on a cosmology-dependent variable) only in Appendix D, as documented pitfalls.

6 Discussion

6.1 An unexplained deficit is not new physics

The surviving anomaly has passed every mundane test we were able to construct: survey geometry, radial selection, redshift space, galaxy density, discretization, field transformation, observational weighting, an HOD refit to the observed clustering, a density-dependent incompleteness surrogate, and the entire $w_0\text{CDM}$ ($w_a = 0$) family. It would nonetheless be a category error to promote it to evidence for physics beyond ΛCDM . The space of unmodelled observational and astrophysical explanations is not exhaustible by enumeration: the official fiber-assignment mocks, photometric and imaging systematics, target-selection effects, assembly bias, and baryonic effects on the tracer field all remain untested here, and a single observed field admits no replication check. The correct reading is: a robust, well-characterised, currently unexplained deviation of the observed field’s loop statistics from its best available ΛCDM forward model — an invitation to explanation, not a detection.

We highlight one physical clue for future work. The deficit co-occurs with a distinctive one-point structure (more concentrated high-density tail, less diffuse small-scale power, Section 5.2) that neither the data-preferred HOD nor incompleteness reproduces. Whatever explains the one-point structure will plausibly explain the topology.

6.2 Implications for topological analyses of surveys

Three practices emerge from this study as, in our view, mandatory for PH on observed fields: (i) report and match σ_{px} , by placing data and mocks on a common grid at common physical smoothing rather than by compensating smoothing scales; (ii) never compare a bounded field analysed in a zero-padded cube against periodic boxes — carve the mocks by the survey geometry and exclude the exterior from the filtration on both sides; (iii) verify the field transformation (Eq. 3.2) is applied, where, and identically on both sides, since amplitude statistics respond to it at the factor-of-two level. More generally, connectivity statistics (β_1^{max}) proved robust to every construction choice we varied, while amplitude statistics ($\langle\text{pers}_1\rangle$) absorbed each residual mismatch in turn; we suggest counting statistics be preferred as primary observables for observational PH.

6.3 Relation to previous work

Our results are complementary to, and do not contradict, the simulation-based PH literature. The forecast advantages of PH over $P(k)$ on periodic boxes [1, 3] concern a regime where none of the systematics characterised here operates; the DES Y3 application [2] works on projected 2D maps, where the boundary and σ_{px} issues take a different form. The RSD sensitivities measured by [4] on simulated fields are consistent with the sub-dominant RSD shifts we find

once RSD is applied on both sides of the comparison. What this paper adds is the missing bridge to observed 3D fields: the systematics that appear only there, and a construction under which the comparison becomes meaningful. Field-level inference programmes such as SimBIG [13–15] face the same forward-modelling requirements; our cut-sky construction is a TDA-specific instance of that general lesson.

7 Limitations and future work

We list the open items explicitly and without softening.

1. **Fiber assignment.** Our incompleteness test is a surrogate model. The definitive test requires the DESI alternate-MTL / fast-fiberassign mock catalogues, which we have not processed. Until then, fiber assignment is disfavoured (wrong sign in the surrogate; weight-insensitivity of $\beta_1^{\max}$) but not excluded.
2. $w_a \neq 0$. The exclusion of Section 5.4 covers w_0 CDM with $w_a = 0$ only. CPL cosmologies — including the DESI DR2 best fit — require dedicated simulations.
3. **Single observed field.** The anomaly rests on one field (with the SGC as an internal cross-check sharing the pipeline and hence its potential blind spots). Independent replication on BOSS DR12 [16] and on DESI DR2, under this frozen protocol, is the natural confirmation path and would also resolve the look-elsewhere concerns inherent to any post-hoc-discovered feature.
4. **Error budget.** The quoted rank does not include cosmic variance at the BGS volume on the data side; volume-matched lightcone suites would be required for a complete significance.
5. **HOD.** The clustering fit is a coarse-grid, fixed-scatter calibration at $z = 0.5$ with the minimum at the grid edge in $\log M_1$; a full five-parameter posterior propagated through the pipeline (and a SHAM-based construction matched to BGS) remains to be done.
6. **Snapshot vs. lightcone.** Mocks are $z = 0.5$ snapshots; the data span $z \in [0.1, 0.4]$. The growth mismatch is common to all mocks and cancels in mock-to-mock comparisons, but contributes an unquantified offset to the data-mock comparison.
7. **Tiling.** Cut-sky mocks tile a $1000 h^{-1}\text{Mpc}$ periodic box into a $1997 h^{-1}\text{Mpc}$ cube, repeating structure at the box scale; standard practice, but a correlated-noise contribution we have not quantified.
8. **Satellite profiles.** Satellites are placed uniformly within the virial radius rather than following an NFW profile; at the smoothing scales used this is sub-dominant but untested.
9. **Fiducial cosmology.** The redshift-to-distance conversion assumes the Quijote fiducial; Alcock–Paczynski sensitivity of the voxelized topology is not assessed.

8 Conclusions

We have presented, to our knowledge, the first application of three-dimensional persistent homology to an observed spectroscopic galaxy field, and we have found that the transition from simulated boxes to observed surveys is dominated by systematics capable of manufacturing high-significance topological anomalies. Our contributions are:

1. **The σ_{px} systematic:** the mean H_1 persistence responds to the smoothing scale in voxel units ($+8.42\sigma$ across the relevant range), so cross-resolution PH comparisons must match σ_{px} by placing both fields on a common grid.
2. **The boundary artefact:** zero-padded embedding plus unmasked filtration truncates the persistence diagram, producing an anti-correlated high-persistence/low-count “dual signature” that mimics a physical signal; our own initial detection was largely of this type, and we document its dissolution.
3. **A like-for-like framework:** cut-sky, selection-matched, redshift-space, density-matched mocks with a masked filtration — the minimal construction under which observed-vs-simulated PH is meaningful — released as a reusable pipeline.
4. **An honest residual:** a $\sim 20\%$ deficit of H_1 generators in DESI BGS NGC relative to all 2000 like-for-like mocks (empirical $p < 1/2000$), robust to weighting, to an HOD refit to the observed clustering, and to an incompleteness surrogate, and explicitly *not* attributable to $w_0\text{CDM}$ ($w_a = 0$), for which no parameter value approaches the observation. We report it as an open anomaly, with the outstanding tests (official fiber-assignment mocks, $w_a \neq 0$ simulations, independent replication) enumerated in Section 7.

The trajectory of this project — from an apparent multi-sigma dark-energy signal to its systematic dissolution and a sharper, humbler residual — is itself the demonstration of the framework. We offer the pipeline, the frozen records, and the documented pitfalls as infrastructure for the community’s next attempts at topological cosmology on real data.

A TDA feature nomenclature

The Branch A pipeline extracts eight scalar features per field, listed in Table 3 with their internal code names as they appear in the public repository. Features at positions 0–3 derive from the β_1 Betti curve (prefix **b1_**); features at positions 4–6 derive from the H_1 persistence diagram and position 7 from the β_0 curve. The internal prefix **b2_** at positions 4–6 is a positional label and does not denote the Betti number β_2 : no H_2 features are computed. A dedicated diagnostic on 10 mock fields showed that H_2 generators at 128^3 resolution are lattice artefacts of the cubical discretization — physical fields produce *fewer* H_2 generators than a smoothed Gaussian random field at identical resolution (physical/GRF ratio $0.62 < 1$) — so H_2 analysis is deferred to higher grid resolution.

B σ_{px} sensitivity: complete numerical data

Table 4 provides the complete numerical characterisation of the σ_{px} sensitivity of $\langle \text{pers}_1 \rangle$ on $N = 200$ Quijote nwLH mock fields, all on the mock grid ($\Delta x = 7.81 h^{-1}\text{Mpc}$) so that only

Table 3. Branch A feature nomenclature. The two features used in this paper are $\beta_1^{\max}$ (index 4) and $\langle \text{pers}_1 \rangle$ (index 5).

Index	Code name	Symbol	Group	Description
0	<code>b1_peak_pos</code>	$\nu_{1,\text{peak}}$	H_1/β_1	threshold of the β_1 peak
1	<code>b1_peak_height</code>	β_1^{peak}	H_1/β_1	β_1 curve maximum
2	<code>b1_fwhm</code>	$\Delta\nu_1$	H_1/β_1	FWHM of the β_1 curve
3	<code>b1_integral</code>	$\int \beta_1$	H_1/β_1	area under the β_1 curve
4	<code>b2_max_count</code>	$\beta_1^{\max}$	H_1/β_1	finite H_1 generator count
5	<code>b2_mean_persistence</code>	$\langle \text{pers}_1 \rangle$	H_1/β_1	mean finite H_1 persistence
6	<code>b2_high_persist</code>	β_1^{high}	H_1/β_1	count above the 90th pers. pct.
7	<code>b0_at_mean</code>	$\beta_0(\bar{\nu})$	H_0/β_0	components at the mean level

σ_{px} varies. The difference between the first two rows — $\Delta\langle \text{pers}_1 \rangle = +0.153$, i.e. +8.42 times the mock scatter at $\sigma_{\text{px}} = 0.640$ — is the systematic quoted in Section 3.2. We deliberately do not tabulate a data-vs-mock z -score against these rows: comparing a field on one grid to a distribution on another is exactly the biased construction that Section 3.2 rules out, and the earlier version of this analysis that did so is documented in Appendix D.

Table 4. $\langle \text{pers}_1 \rangle$ as a function of $\sigma_{\text{px}} = R/\Delta x$ on $N = 200$ Quijote nwLH mock fields (mock grid, $\Delta x = 7.81 h^{-1} \text{Mpc}$).

$R (h^{-1} \text{Mpc})$	σ_{px}	Mock mean	Mock std
1.7	0.216	0.106	0.005
5.0	0.640	0.238	0.019
10.0	1.280	0.394	0.015

C Cut-sky mock construction: technical details

This appendix specifies the cut-sky construction of Section 4 at the level required for independent reimplementations; the reference implementation is in the public repository.

Halo catalogues and velocities. Friends-of-Friends catalogues are read from the Quijote `group_tab` binaries at snapshot $z = 0.5$, including the group bulk velocities, which are converted to peculiar velocities with the Gadget $\sqrt{a}$ convention ($a = 2/3$ at $z = 0.5$). Validation on a reference simulation gives $\langle |v_h| \rangle = 356 \text{ km s}^{-1}$ with a maximum of 1540 km s^{-1} , consistent with halo peculiar velocities at this redshift. Haloes with $N_{\text{part}} \geq 20$ ($M_h \gtrsim 8.6 \times 10^{12} h^{-1} M_\odot$) are retained.

HOD population. The Zheng et al. [11] occupation is sampled with baseline parameters $(\log M_{\min}, \sigma_{\log M}, \log M_0, \log M_1, \alpha) = (12.5, 0.55, 12.25, 13.5, 1.0)$ and, for the robustness test of Section 5.3, with the clustering-fitted values of Table 2. Centrals sit at the halo centre with the halo bulk velocity; satellites are placed uniformly within the virial radius ($r = R_{\text{vir}} u^{1/3}$, isotropic angles) and receive the halo bulk velocity plus an isotropic Gaussian dispersion of one-dimensional width $\sigma_{1D} = \sqrt{GM_h/3R_{\text{vir}}}$.

Tiling, observer, and redshift space. The periodic box is replicated over all integer offsets intersecting the DESI embedding cube (box minimum and side taken verbatim from the data voxelization: $L = 1997.36 h^{-1}\text{Mpc}$, $\Delta x = 15.604 h^{-1}\text{Mpc}$). The observer sits at the Cartesian origin of the data coordinate system; comoving distance yields z_{cosmo} by inverse interpolation, and $z_{\text{obs}} = z_{\text{cosmo}} + (1 + z_{\text{cosmo}}) v_{\parallel}/c$ with the galaxy repositioned at the comoving distance of z_{obs} along its line of sight. Galaxies are kept if $z_{\text{obs}} \in [0.1, 0.4]$ and their embedding voxel lies in the DESI NGC mask (defined as voxels whose random-catalogue CIC density exceeds 1% of its mean — the same criterion used for the data).

Selection matching. Candidates are downsampled radially with a bin-wise acceptance probability proportional to the BGS $n(z)$ shape and globally normalised so that the expected selected count equals the BGS NGC total of 217,614; the achieved median across the suite is 217,570.

Field and filtration. The selected mock galaxies are CIC-assigned with unit weights and divided against the *same* DESI random field via Eq. (3.1); the log-transform of Eq. (3.2) is applied to the raw contrast; the field is smoothed at $R = 5 h^{-1}\text{Mpc}$ on the DESI grid ($\sigma_{\text{px}} = 0.320$) and mean-subtracted in the mask. After the transform, the in-survey standard-deviation ratio between mock and data fields is 0.79 (it is ~ 31 without the transform, driven by FKP shot-noise spikes up to $\delta \sim +150$ in low-coverage voxels). In the masked filtration, exterior cells receive a sentinel value entering the superlevel filtration last, and any generator whose birth or death reaches the sentinel is discarded; the identical rule is applied to the data field.

D Methodological evolution of the comparison

This project progressed through a sequence of increasingly strict comparisons, and we document the sequence because each stage isolated a systematic of general interest. The progression was driven throughout by a fixed protocol — pre-registered gate thresholds, frozen numerical records, and independent adversarial review of each phase — and it is this protocol, rather than any single result, that we consider the project’s methodological core.

1. *Initial configuration (superseded).* The DESI field, embedded in a zero-padded cube, was compared via an unmasked filtration against periodic-box mocks, at fixed physical R across different grids. This produced $\langle \text{pers}_1 \rangle$ anomalies of $+5.96\sigma$ to $+9.45\sigma$ depending on HOD calibration, a generator-count suppression of $\sim -56\sigma$, and an apparent joint w_0 -consistent signature. A density-controlled subset, selected post hoc on a cosmology-dependent variable, reached $+29.9\sigma$; we identify that construction plainly as a statistical error (conditioning on N_{gal} mechanically shrinks the denominator) and it should not be used. These numbers are superseded and appear here only as documented pitfalls.
2. *σ_{px} characterisation.* Investigating the smoothing sensitivity revealed the grid-unit dependence of Section 3.2, reframing the cross-grid comparison as structurally biased and motivating the common-grid construction.
3. *Restoring the stated transform.* The pipeline audit triggered by the like-for-like rebuild found that the log-transform of Eq. (3.2), stated in the method description, was not applied in the computation; restoring it (in the correct position, on the raw contrast)

shifted the data-side $\langle \text{pers}_1 \rangle$ by a factor ~ 2 and exposed the amplitude fragility of persistence means. The corrected pipeline is the one used throughout this paper.

4. *Cut-sky, then masked.* Carving the mocks by the survey geometry (exact-replica test) removed most of the generator-count suppression; excluding the exterior from the filtration on both sides (masked test) removed the remaining boundary generators and, with them, the significance of the $\langle \text{pers}_1 \rangle$ anomaly. The β_1^{max} deficit alone survived, at every stage of the sequence, with stable magnitude.
5. *Exclusion battery.* Weighting, the clustering-fitted HOD, the incompleteness surrogate, and the w_0 scan (Sections 5.3–5.4) then removed the mundane and dark-energy candidates accessible to us.

The initial detection claim did not survive this sequence; the tools built to test it did, and they are the content of this paper.

E Complementary neural-network null (Branch B)

In parallel with the topological pipeline, the project developed a complementary branch applying an SE(3)-equivariant graph neural network (e3nn [17]) to a latent “tension field” built from the same voxelized inputs, with the partial correlation $r(\mathbf{j}^*, w_0 | \Omega_m, \sigma_8)$ as the target statistic. The final result is a null: $\sigma_B = 1.68 \pm 0.43$ across ten independent training seeds, with the sign of the observed correlation alternating (5 positive, 5 negative) — no stable w_0 direction in the latent space. This is the physically expected outcome for snapshot fields, whose linear growth carries no residual w_0 information once Ω_m and σ_8 are conditioned on, and it is consistent with the w_0 insensitivity of β_1^{max} established in Section 5.4.

The null was reached through five pipeline versions whose intermediate, inflated sensitivities (8.3σ , 5.6σ , 11.8σ) each traced to an identifiable defect (wrong checkpoint, marginal rather than partial correlation, a parameter-column indexing error); each methodologically motivated correction moved the result toward the final null. We retain this history in the spirit of Appendix D.

One caution supersedes our earlier reading of this branch. We previously argued that the independence of the two branches excluded shared artefacts as the origin of the topological signal. That claim was overdrawn: the branches share exactly the upstream stages — catalogue construction, weighting, coordinate conversion, voxelization, and the (non-)treatment of the survey mask — where the systematics of Sections 3.2–3.3 live. Branch independence excludes shared *topological-computation* artefacts only. In retrospect this is fully consistent with the outcome of the like-for-like analysis, which located the dissolved signals precisely in those shared upstream stages.

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AI-assistance disclosure. Portions of this manuscript were drafted with the assistance of a large language model (Claude, Anthropic). All scientific content, numerical results, physical interpretations, and conclusions were conceived, verified, and approved by the author. No AI tool was used to generate data, perform the scientific calculations, or make scientific judgments; its role was limited to drafting and editing text and code under the author’s direction.

Competing interests. The author declares no competing interests.

Data availability. The DESI BGS DR1 catalogues are publicly available through the DESI Data Release portal. The Quijote nwLH simulation suite is publicly available through the Quijote project. The CAUCHY analysis pipeline, including all scripts, frozen numerical records, and gate protocols required to reproduce the results and figures, is available at <https://github.com/AleMarco1/CAUCHY> and archived at Zenodo (DOI: 10.5281/zenodo.21128857).

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